

HR Analytics Dashboard

Introduction

Employee attrition affects organizational productivity and workforce stability. This project analyzes HR employee data using Microsoft Excel and Microsoft Power BI to identify workforce patterns, attrition trends, and support HR decision-making through an interactive dashboard.

Objectives

- Analyze employee data based on salary, department, experience, age and education.
- Identify employee attrition patterns.
- Compare attrition across departments and job roles.
- Perform correlation analysis to identify key factors affecting attrition.
- Build an interactive HR Analytics dashboard.

Tools & Technologies

Microsoft Excel, Power BI, Power Query, DAX

Dataset Information

Dataset: IBM HR Analytics Employee Attrition Dataset

Total Employees: 1,470

Data Cleaning: Removed duplicates, handled missing values, verified data types.

Methodology

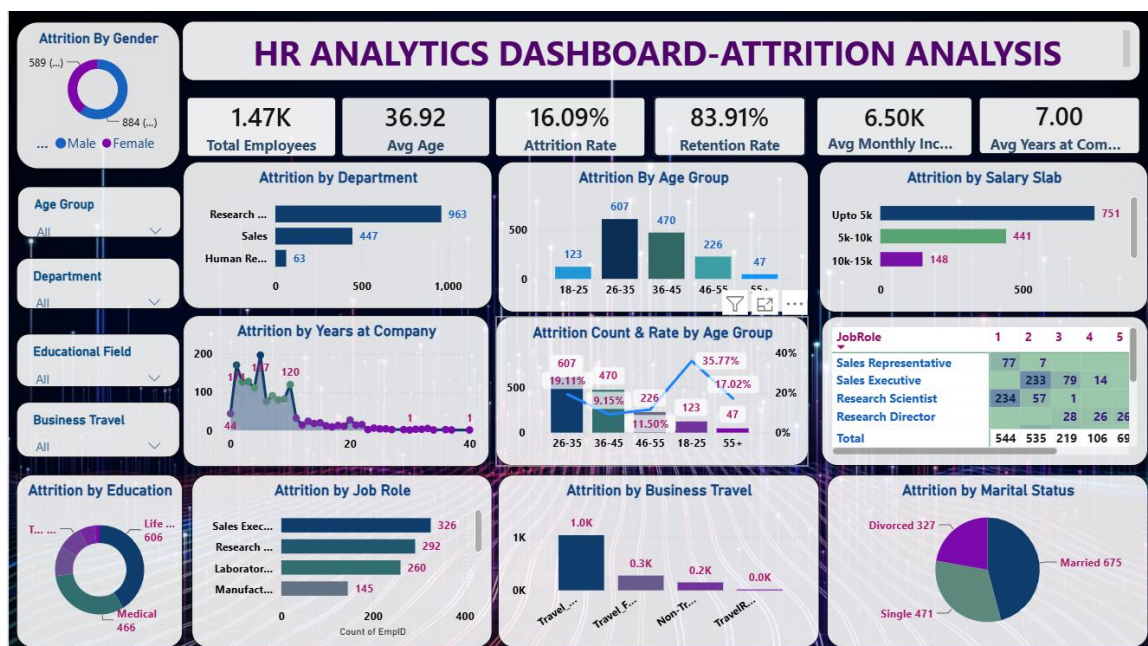
1. Data Collection
2. Data Cleaning
3. Data Transformation
4. KPI Creation
5. Correlation Analysis in Excel
6. Dashboard Development in Power BI
7. Insight Generation

KPI Summary

KPI	Value
Total Employees	1.47K (1470)
Average Age	36.92 Years
Attrition Rate	16.09%
Retention Rate	83.91%
Average Monthly Income	6.50K
Average Years at Company	7.00 Years

Insights

- Research & Development has the highest workforce (963 employees), followed by Sales (447) and Human Resources (63).
- Employees aged 26–35 years form the largest group (607 employees).
- The highest attrition rate is observed in the 18–25 age group (35.77%).
- Most employees belong to the Up to 5K salary slab (751 employees).
- Sales Executive (326) and Research Scientist (292) are the largest job roles.
- Life Sciences (606) is the leading education field.
- Married employees (673) represent the largest marital-status group.
- Most employees travel rarely for business.



Correlation Analysis

- Job Level ↔ Monthly Income: Strong positive correlation (0.95).
- Job Level ↔ Total Working Years: Strong positive correlation (0.78).
- Monthly Income ↔ Total Working Years: Strong positive correlation (0.77).
- Years at Company ↔ Years in Current Role: Strong positive correlation (0.76).
- Performance Rating ↔ Percent Salary Hike: Strong positive correlation (0.77).

Recommendations

- Implement targeted retention programs for employees aged 18–25 years.
- Provide career growth and mentoring opportunities for early-career employees.
- Review compensation and benefits for lower salary slabs.
- Conduct regular employee satisfaction surveys.
- Strengthen training and development initiatives.
- Improve work-life balance through flexible HR policies.
- Monitor attrition trends regularly using the Power BI dashboard.

Conclusion

The HR Analytics Dashboard provides meaningful insights into employee demographics, salary distribution, department-wise workforce, and attrition patterns. The dashboard enables HR teams to monitor KPIs and make data-driven decisions to improve employee retention and workforce planning.